

Rising Waters

Flood Management System — AI/ML Project

Sample Project Documentation

Phase 7: Project Documentation

IBM Cloud

XGBoost 96.55%

Skill Wallet

AI/ML Track

1. Project Title

Rising Waters — Machine Learning Approach to Flood Prediction

2. Problem Statement

Floods are among the most devastating natural disasters, claiming thousands of lives annually. Conventional forecasting methods lack the predictive intelligence required for timely early warnings. This project addresses this gap by building an ML-powered flood prediction web application.

3. Objective

- Train and compare 4 ML classifiers on meteorological flood data
- Deploy the best model (XGBoost) as a Flask web application
- Provide disaster management officers with real-time flood risk predictions
- Deliver dedicated result pages for Flood Chance and No Flood Chance

4. Dataset

Size	5,000 records
Features	annual_rainfall, seasonal_rainfall, cloud_visibility, humidity, temperature, river_level
Target	flood_occurred (0=No Flood, 1=Flood)
Source	Synthetic dataset based on real meteorological patterns

5. Model Performance

Decision Tree	78.17% accuracy
K-Nearest Neighbours	80.67% accuracy
Random Forest	82.00% accuracy
XGBoost (DEPLOYED)	96.55% accuracy — saved as floods.save

6. Web Application

Home Page	Dashboard with model stats and prediction counts
Prediction Page	6-field meteorological input form
Flood Result Page	Red danger banner, alert level, 6 emergency actions
No Flood Result Page	Green safe banner, all-clear badge, monitoring tips
History Page	Full audit log of all past predictions
About Page	Architecture, scenarios, technology stack

7. Deployment

The application is deployed on IBM Cloud using Docker containerization and a Cloud Foundry manifest.yml. It is accessible 24/7 via a public cloud URL.

8. Project Structure

app.py	Flask application controller
model/train_model.py	Complete ML training pipeline
model/floods.save	Deployed XGBoost model
templates/	6 Jinja2 HTML templates
static/css/style.css	Premium dark-theme CSS
utils/preprocessing.py	Input validation and scaling
requirements.txt	Python dependencies
Dockerfile	Container config
manifest.yml	IBM Cloud manifest